

C - Programming

Name: N. Pratik Raj

Reg No: 192312284

11/08/26

CO3 - AT:2

Course Code: CSA0209

Q1) Write a Factorial program by using Recursive Function

Ans Code :

```
#include <stdio.h>
long long factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    }
    else {
        return n * factorial(n-1);
    }
}
```

Syntax

```
returntype Factorial (data type  
n)  
{  
    if (condition)  
        return value;  
    else  
        return n * factorial  
        (n-1);  
}
```

```
int main () {
    int num;
    printf ("Enter a number: \n");
    scanf ("%d", &num);
    if (num < 0) {
        printf ("Factorial of %d is not defined for negative  
numbers. \n");
    }
    else {
        printf ("Factorial of %d = %.lld \n", num, factorial  
(num));
    }
    return 0;
}
```

Sample Output : Enter a number = 5.
Factorial of 5 = 120.